Matthew Spitzer Brooks

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Education

Ph.D. Agricultural and Resource Economics, University of California, Davis	Expected June 2027
M.S. Agricultural and Resource Economics, University of California, Davis	2024
B.A. Economics and Religion, *summa cum laude*, Phi Beta Kappa, Middlebury College	2017

References

Katrina Jessoe **(Chair)**
Associate Professor
Department of Agricultural and Resource Economics
University of California, Davis
kkjessoe@ucdavis.edu

Travis Lybbert
Professor
Department of Agricultural and Resource Economics
University of California, Davis
tlybbert@ucdavis.edu

Jamie Hansen-Lewis
Assistant Professor
Department of Agricultural and Resource Economics
University of California, Davis
jhansenlewis@ucdavis.edu

Research Fields

Environmental Economics, Development Economics, Health Economics

Research and Work In Progress

The Pollution–Productivity Curve:
Non-Linear Effects and Adaptation in High-Pollution Environments (**Job Market Paper**) Link to PDF
with F. Usmani, Global Energy Alliance for People and Planet

Abstract. This paper studies whether workers adapt to air pollution using ball-by-ball data from Indian cricket, where match schedules and training locations generate variation in exposure. A 10 $\mu g\,m^{-3}$ increase in match-day $PM_{2.5}$ reduces bowlers' performance relative to batters' by about 1 percent. This effect is statistically significant only in the top pollution quintile (over 5 times the U.S. annual average), although we cannot reject a linear dose–response. Higher exposure in the prior month attenuates this effect, consistent with adaptation, but lowers performance by more than attenuation offsets. Results suggest lowering baseline pollution, not merely averting extremes, yields productivity gains.

Strengthening Women's Livestock Cooperatives in Nepal with an Innovative Technology Bundle
with T. Lybbert (UC Davis), N. Magnan (Colorado State U), C. Mullally (U Florida), and B. Shrestha (Heifer)

Monetary and Non-monetary Barriers to Accessing Environmental Public Benefit Programs: Experimental Evidence from California
with S. Nakamura (Pennsylvania State U) and C. Weigel (California Air Resources Board)

Partisan Differences in Adaptation to Wildfire Smoke: Evidence from Indoor Air Quality in California
with J. Hansen-Lewis (UC Davis)

Global Spillovers in Agricultural Technology Development
with A. Shenoy (UC Davis)

Publications

Peer-reviewed:

- "A Taste for Taxes: Minimizing Distortions Using Political Preferences." 2019. ***Journal of Public Economics***, vol. 180, p. 104055. With Andrea Robbett and Emiliano Huet-Vaughn. Link.

Policy Publications:

- "Does An Innovation's Reach Reveal Anything About Its Impact? Under The Right Conditions: Possibly." 2025. Rome, Italy: Standing Panel on Impact Assessment. With Travis Lybbert and Madeleine Walker. Link to PDF.
- "Evaluation of the Liberia Compact's Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Findings from the Final Round." 2024. Washington, DC: Mathematica. With Candace Miller, et al. Link to PDF.
- "Liberia Energy Project: Findings from the Final Evaluation of the Pipeline Sub-Activity." 2024. Washington, DC: Mathematica. With Poonam Ravindranath, Paolo Abarcar, Cullen Seaton, and Candace Miller. Link to PDF.
- "Maternal Health Care Quality Improvement in Rajasthan, India: Insights from a Development Impact Bond Verification Agent." 2021. Washington, DC: Mathematica. With So O'Neil, Divya Vohra, Shveta Kalyanwala, and Dana Rotz. Link to PDF.
- "Jordan Refugee Livelihoods Development Impact Bond Evaluation Framework." 2021. Washington, DC: Mathematica. With Evan Borkum, Paolo Abarcar, and Laura Meyer. Link to PDF.
- "Maternal Health Care Quality and Outcomes Under the Utkrisht Impact Bond: Midline Findings and Insights." 2020. Washington, DC: Mathematica. With So O'Neil, Divya Vohra, Dana Rotz, and Shveta Kalyanwala. Link to PDF.
- "Evaluation of the Liberia Power Compact's Mt. Coffee Hydropower Plant Rehabilitation and Capacity Building and Sector Reform: Baseline and Interim Findings." 2020. Washington, DC: Mathematica. With Candace Miller, et al. Link to PDF.

Research Experience

Graduate Student Researcher
University of California, Davis
with Dr. Travis Lybbert 2024–present
with Dr. Ashish Shenoy 2023–2024
Research Analyst
Mathematica Policy Research, Cambridge, MA 2018–2022
Policy and Communications Associate
Innovations for Poverty Action, New Haven, CT 2017–2018
Research Assistant
Princeton University, Princeton, NJ and
Busara Center for Behavioral Economics, Nairobi, Kenya
with Dr. Jeremy P. Shapiro 2016–2017

Fellowships, Honors, and Awards

J-PAL Full-RCT grant long-term impacts top up (collaborator), $50,902	2026
Henry A. Jastro Graduate Research Award, $3,000, UC Davis	2026
ARE Graduate Summer Research Fellowship, $3,000, UC Davis	2026
Graduate Studies Travel Award, $1,500, UC Davis	2026
Henry A. Jastro Graduate Research Award, $3,000, UC Davis	2025
ARE Graduate Travel Award, UC Davis	2025
SurveyCTO Primary Data Collection Research Grant Honorable Mention	2024
J-PAL Full-RCT grant (collaborator), $375,000	2024
Henry A. Jastro Graduate Research Award, $3,000, UC Davis	2024
ARE Graduate Summer Research Fellowship, $4,000, UC Davis	2024
Provost's Fellowship in the Arts, Humanities and Social Sciences, $\approx \$59,000$, UC Davis	2022
Phi Beta Kappa, Middlebury College	2017
Highest Honors in Economics, Middlebury College	2017
Highest Honors in Religion, Middlebury College	2017
Religion Departmental Prize, Middlebury College	2017
Dirks Award (Economics senior of high promise), Nominee, Middlebury College	2017
Fulbright Study/Research Grant Nepal, Semi-Finalist	2017
Thomas J. Watson Fellowship, Semi-Finalist	2017
Kellogg Fellow, $5,000 grant for senior thesis, Middlebury College	2016
College Scholar, Middlebury College	2013-2017

Professional Activities

Invited Seminars

Middlebury College, Department of Economics	Jun. 2026
University of Massachusetts - Amherst, Department of Resource Economics	Sep. 2025
Indian Statistical Institute, Delhi	Jul. 2025
Centre for Research on the Economics of Climate, Food, Energy, and Environment	

Conference Presentations

Association for Public Policy Analysis & Management Fall Research Conference (scheduled)	Nov. 2026
Center for Effective Global Action Research Retreat, UC Berkeley	Oct. 2026
Liberal Arts Colleges Development Economics Conference (LACDev; attendee), Wesleyan	Sep. 2026
Agricultural and Applied Economics Association Annual Meeting (4 presentations)	Jul. 2026
World Congress of Environmental and Resource Economists, Portugal	Jun. 2026
Giannini Foundation of Agricultural and Resource Economics Student Conference, UC Berkeley	May 2026
Pacific Conference for Development Economics (PacDev) at UC Davis	Mar. 2026
Occasional Workshop in Environmental & Resource Economics at UC Santa Barbara	Oct. 2025
Agricultural and Applied Economics Association Annual Meeting	Jul. 2025

Referee Service

American Journal of Agricultural Economics, *Energy Economics*, *PLOS One*

Departmental Service

Member of Principals of the Community Committee	2025–2026
Teaching Assistant Coordinator	2025–2026
Graduate Student Association Representative (Alternate)	2024–2026
Graduate Student Association Representative	2023-2024

Teaching Experience

Instructor

Applied Economics Research Course (UC Davis, undergraduate) — Summer 2026

- A five-week course mentoring students on independent research. Emphasis on designing a research question, working with data, and applying econometric techniques.

- Prerequisite: ARE 106 — Econometric Theory & Applications.

Teaching Assistant

ARE 133: Introduction to Behavioral Economics (UC Davis, undergraduate)	Fall 2026
ARE 115A: Economic Development (UC Davis, undergraduate)	Winter 2026
ARE 100B: Intermediate Microeconomics (UC Davis, undergraduate)	Winter 2025

Guest Lecturer

ARE 133: Introduction to Behavioral Economics (UC Davis, undergraduate)	April 7, 2026
ARE 176: Environmental Economics (UC Davis, undergraduate)	December 3, 2025

Mentoring

Diversity and Inclusion in Research, Education, and Career Training Program (UC Davis)	
Student Café Graduate Student Coordinator	2026
Graduate Inclusive Education and Mentoring Training (GET) Workshop	2025
Tutor for Undergraduates in Department of Agricultural and Resource Economics Courses	2025

Additional

Technical skills:
R, Stata, Python, MATLAB, Julia, Google Earth Engine; LaTeX
In-person fieldwork:
Nepal, Kenya, Liberia
Citizenship
U.S. Citizen